Mohab Kishawy

Ph.D. Candidate in Electrical and Computer Engineering, McMaster University | Available January 2027

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SUMMARY

Computer vision researcher working on physically grounded image restoration and 3D scene reconstruction, with first-author papers at ICPR 2026, the CVPR 2026 Workshops, and *Scientific Reports*, and four top-5 finishes in NTIRE challenges at CVPR (2024–2026). Seeking research scientist roles in computer vision and 3D vision from January 2027 (onsite or remote).

EDUCATION

Ph.D. in Electrical and Computer Engineering — Sep 2023 – Dec 2026 (expected)
McMaster University — Hamilton, ON, Canada

- Thesis: *Modelling the Physics of Degradation: Retinex-Based Restoration from 2D Images to 3D Scenes*
- Advisor: Prof. Jun Chen | GPA: 11.0/12.0 | Full scholarship

B.Sc. in Computer Engineering — Sep 2018 – May 2023
Nile University — Giza, Egypt

- Thesis: *Federated Learning System on Autonomous Vehicles for Lane Segmentation* (published in *Scientific Reports*)
- Advisor: Dr. Mohamed S. Darweesh | GPA: 3.7/4.0 | Full scholarship

RESEARCH EXPERIENCE

Computer Vision Researcher — Sep 2023 – Present
McMaster University, advised by Prof. Jun Chen — Hamilton, ON, Canada

- Physically grounded, Retinex-based restoration of ultra-high-definition images (Mamba, frequency-domain modelling) and of 3D scenes (3D Gaussian Splatting); released open-source code and pre-trained models.
- Led team entries in five NTIRE challenges at CVPR (2024, 2026) and competed in a sixth (2025), with four top-5 finishes.

Machine Learning Researcher — Dec 2021 – Dec 2022
Helwan University, Electronics and Computer Engineering Lab — Cairo, Egypt

- Automated assessment of students in online labs from mouse-tracking data, saving 32 hours of grading per term.

Research Assistant — Feb 2021 – Aug 2021
Nile University, Wireless Intelligent Network Center (WINC) — Giza, Egypt

- Plant-disease detection with a modified CNN; federated learning for lane segmentation on autonomous vehicles.

PUBLICATIONS

Earlier work appears under the name Mohab M. Eid. Full list with links: errorlogic1211.github.io/publications

Selected publications

Under review — **M. Kishawy**, C. Tang, S. Shirani, J. Chen. **VeilGS: Solving Darkness as a Veil of 2D Basis Maps over Retinex 3DGS**. Manuscript available on request.

ICPR '26 — **M. Kishawy**, A. A. Hussein, J. Chen. **RetinexDual: Retinex-Based Dual Nature Approach for Generalized Ultra-High-Definition Image Restoration**. International Conference on Pattern Recognition (ICPR), LNCS, Springer, pp. 90–105, 2026. Poster. [Paper] [Code]

CVPRW '26 — **M. Kishawy**, J. Chen. **RetinexDualV2: Physically-Grounded Dual Retinex for Generalized UHD Image Restoration**. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 1989–1998, 2026. Poster. [Paper] [Code]

Sci. Rep. '24 — **M. M. Eid Kishawy**, M. T. Abd El-Hafez, R. Yousri, M. S. Darweesh. **Federated Learning System on Autonomous Vehicles for Lane Segmentation**. *Scientific Reports*, vol. 14, art. 25029, 2024. [Paper] [Code]

Other publications (NTIRE reports are co-authored as a participating team; our placement in brackets)

CVPRW '26 — X. Li, Y. Jin, S. Yao, …, **M. Kishawy**, J. Chen, … (98 authors). *NTIRE 2026 The Second Challenge on Day and Night Raindrop Removal for Dual-Focused Images: Methods and Results*. CVPR Workshops, pp. 2054–2070, 2026. **[4th place]**

CVPRW '26 — G. Ciubotariu, S. S. M. A, …, **M. Kishawy**, J. Chen, … (91 authors). *Low Light Image Enhancement Challenge at NTIRE 2026*. CVPR Workshops, pp. 1741–1753, 2026. **[5th place, Track 2: Joint Denoising and Low-Light Enhancement]**

CVPRW '26 — R. P. Ancuti, A. Brateanu, …, **M. Kishawy**, … (89 authors). *NTIRE 2026 Nighttime Image Dehazing Challenge Report*. CVPR Workshops, pp. 1608–1625, 2026.

CVPRW '26 F.-A. Vasluianu, T. Seizinger, …, **M. Kishawy**, J. Chen, … (90 authors). *Advances in Single-Image Shadow Removal: Results from the NTIRE 2026 Challenge*. CVPR Workshops, pp. 2582–2595, 2026.

CVPRW '25 X. Liu, Z. Wu, F.-A. Vasluianu, …, **M. Kishawy**, J. Chen, … (105 authors). *NTIRE 2025 Challenge on Low Light Image Enhancement: Methods and Results*. CVPR Workshops, pp. 1196–1206, 2025. **[2nd place; highest PSNR]**

CVPRW '24 C. O. Ancuti, C. Ancuti, …, **M. Kishawy**, … (68 authors). *NTIRE 2024 Dense and Non-Homogeneous Dehazing Challenge Report*. CVPR Workshops, pp. 6453–6468, 2024. **[4th place]**

IEEE Access '22 A. M. Abd El-Haleem, **M. M. Eid**, M. M. Elmesalawy, H. A. H. Hosny. *A Generic AI-Based Technique for Assessing Student Performance in Conducting Online Virtual and Remote Controlled Laboratories*. *IEEE Access* 10:128046–128065, 2022. [Code]

CICN '22 H. A. Hassan, **M. M. Eid**, M. M. Elmesalawy, A. M. Abd El-Haleem. *A New Intelligent System for Evaluating and Assisting Students in Laboratory Learning Management System*. CICN, pp. 566–571, 2022. [Code]

IJAMC '21 M. M. El-Lamey, **M. M. Eid**, M. Gamal, N. M. Bishady, A. W. Mohamed. *Using Machine Learning Algorithms for Breast Cancer Diagnosis*. *Int. J. of Applied Metaheuristic Computing*, vol. 12, no. 4, pp. 117–137, 2021. Reprinted in *Research Anthology on Medical Informatics in Breast and Cervical Cancer*, IGI Global, 2022.

Springer '21 A. M. Ibrahim, N. E. M. Bishady, S. H. Elghalban, N. N. Mostafa, **M. M. Eid**, K. M. Abdelhamid. *Modelling and Analysis the Population Density Effect on the Infectiousness Rate of COVID-19 Novel Virus and the Mortality Rate Percentage Regarding Oxford's Stringency Index Model of Governmental Response in MATLAB*. In *Artificial Intelligence for COVID-19*, Springer, pp. 437–462, 2021.

NILES '20 A. M. Ibrahim, **M. M. Eid**, N. N. Mostafa, N. E. M. Bishady, S. H. Elghalban. *Modeling the Effect of Population Density on Controlling COVID-19 Initial Spread with the Use of MATLAB Numerical Methods and Stringency Index Model*. Novel Intelligent and Leading Emerging Sciences Conference (NILES), pp. 612–617, 2020.

HONORS AND AWARDS

2026 **4th Place**, NTIRE 2026 Day and Night Raindrop Removal Challenge, CVPR 2026
5th Place, NTIRE 2026 Joint Denoising and Low-Light Image Enhancement (JDLLIE) track, CVPR 2026
2025 **2nd Place**, NTIRE 2025 Low Light Image Enhancement Challenge, CVPR 2025 (highest PSNR)
2024 **4th Place**, NTIRE 2024 Dense and Non-Homogeneous Dehazing Challenge, CVPR 2024
2023 **1st Place**, Hacktrick AI Hackathon, Dell Technologies
2nd Place, AIC Competition for Arabic Text Summarization, Military Technical College, Egypt
Full Ph.D. Scholarship, McMaster University (2023–2026)
2022 **Best Solution Challenge Chapter (Global)**, Google Developer Student Clubs
Shield of Contribution, School of Engineering and Applied Sciences, Nile University
2021 **Dean's Honor List** (GPA above 3.75/4.00, Fall and Spring), Nile University
2018 **Full B.Sc. Scholarship**, Nile University (2018–2023)
2017 **3rd Place (Embedded Systems)**, Intel International Science and Engineering Fair (ISEF)
2015 **Bronze Medal**, Egyptian Olympiad in Informatics

TEACHING AND SERVICE

Teaching Assistant, McMaster University Sep 2023 – Present

COMPENG 2SH4 Principles of Programming (Fall 2025, Fall 2026) | COMPENG 3SM4 Algorithm Design and Analysis (Winter 2024, 2025, 2026) | ELECENG 3TQ3 Advanced Probability and Random Processes (Fall 2024) | COMPENG 4DM4 Computer Architecture (Fall 2023)

Reviewer: AAAI Conference on Artificial Intelligence (AAAI 2025); *IEEE Access* (6 reviews, 2022–2023); *Entropy* (2026)

SKILLS

Programming	Python, C/C++, MATLAB, Java, Bash
Deep learning	PyTorch, CUDA, TensorFlow/Keras, Hugging Face, timm, BasicSR, mamba-ssm, W&B, multi-GPU training
Vision and 3D	Image restoration and enhancement, 3D Gaussian Splatting, NeRF, OpenCV, segmentation (U-Net), YOLO, SAM
Other	Federated learning (Flower), NVIDIA Jetson/TensorRT, Git, Linux, LaTeX
Languages	English (fluent), Arabic (native)

LEADERSHIP AND OUTREACH

At Nile University: **Lead**, Google Developer Student Clubs (2021–2022; 500+ students, workshops and hackathons) | **Deep Learning Instructor**, Root Student Club (2021; 6-month bootcamp, 120+ students) | **Senator**, Student Union (2019–2021) | **ICPC Instructor** (2019–2020)